

# Browser Bundle RC22

## User Manual

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Version 0.2.0 rc22 parallel image preview

**Browser sRGB → PKL FULL REFERENCE → 4C / 7C ECG**

This manual explains how to select and freeze ATLAS colour identities, build portable palettes, prepare independent 4C and ECG paths, create local ICC previews and export traceable evidence.

**Important.** RC22 is a release candidate. Its validation record is READY\_PENDING\_VISUAL\_AUDIT. A screen preview is not a measured print, contract proof or production approval.

**Every colour has an address.**

# 01 Quick start

ATLAS Clarus runs locally in a modern browser. No account, server or network connection is required after download.

## 1 Extract the ZIP file

Do not run the application from inside the ZIP archive. Extract the complete folder and keep its internal structure unchanged.

## 2 Open index.html

Open atlas-clarus-browser-bundle/index.html in a current desktop browser. The header should show OFFLINE · LOCAL.

## 3 Choose a colour source

Use Image Picker for an image pixel or Hover Library for a known ATLAS reference.

## 4 Freeze and save the identity

Add the selected reference to a palette and export Clarus JSON. Browser storage is convenient but is not a backup.

## 5 Continue to production preparation

Open Prepare for print, attach the appropriate CMYK and/or 7CLR output profile, record the settings and export the handoff JSON.

### MAP Main areas

| Area            | Purpose                                                              |
|-----------------|----------------------------------------------------------------------|
| Image Picker    | Resolve an observed image pixel or area mean to one ATLAS reference. |
| Hover Library   | Search and inspect all 13,283 frozen references.                     |
| Identity Wheel  | Explore relationships without changing the selected identity.        |
| Print handoff   | Prepare independent 4C and ECG paths and local ICC previews.         |
| Appearance      | Explore illustrative material and viewing effects.                   |
| App connections | Read guidance for external design applications.                      |
| FAQ and About   | Review claim boundaries, validation and licensing.                   |

## 02 Understand the identity model

The source colour is resolved to the active PKL master before any production path is prepared. The ATLAS address remains the reference identity. Output profiles create device-specific preparations; they do not rewrite the master.

| Layer              | Meaning                                                  | Not a claim of                           |
|--------------------|----------------------------------------------------------|------------------------------------------|
| Observed source    | RGB pixels read from a local image.                      | Spectral measurement or physical colour. |
| PKL Full Reference | Frozen address, atlas_row_id, master RGB and provenance. | Universal device reproduction.           |
| 4C path            | Preparation bound to a CMYK output profile.              | Measured or certified print.             |
| ECG path           | Preparation bound to a seven-channel 7CLR profile.       | Automatic press compatibility.           |
| Measured QC        | A later result appended to the record.                   | Permission to overwrite identity.        |

**Rule.** Keep identity, production preparation and physical measurement as separate layers.

### CHECK Validation baseline

- ✓ Reference count: 13,283.
- ✓ Master SHA-256: 8283ab91b10f89ac758d09ecf5fb4d6343536600a06dd468b1cc1ecf4ec747c4.
- ✓ atlas\_row\_id values are zero-based and unique.
- ✓ A' v0.4 selection logic is unchanged; v0.5 Wheel values are diagnostic only.
- ! Status: READY\_PENDING\_VISUAL\_AUDIT.

# 03 Use the Image Picker

The Image Picker is the shortest route from an image to a documented ATLAS reference. Image data is processed locally.

## 1 Choose an image

Load PNG, JPEG, WebP or GIF. Production preview accepts PNG, JPEG or WebP.

## 2 Choose sampling

Single pixel is exact for one pixel. Area mean  $5 \times 5$ ,  $11 \times 11$  or  $21 \times 21$  is more stable for textured regions.

## 3 Click the intended region

Use the loupe. Pixels with alpha below 128 are excluded; edge samples are clipped.

## 4 Review the result

Check reference, atlas\_row\_id, master RGB, HEX and provenance.

## 5 Add or continue

Add to the palette, open in the Wheel, create a reference card or prepare for print.

## EXAMPLE Example

A designer selects an orange flower petal. The browser resolves the observed sRGB sample to one frozen ATLAS address. Subsequent 4C and ECG previews start from this same identity; neither route may redefine it.

| Sampling       | Best use                                 |
|----------------|------------------------------------------|
| Single pixel   | Flat digital artwork and uniform fields. |
| $5 \times 5$   | Small photographic details.              |
| $11 \times 11$ | General photographic regions.            |
| $21 \times 21$ | Broad textures away from edges.          |

**Boundary.** Area averaging is not spectral measurement, physical equality, production approval or measured QC.

# 04 Find references and build palettes

## SEARCH Search the Hover Library

Search by HLC reference, atlas\_row\_id or HEX. Use the lightness filter and page controls. Selecting a card opens its frozen identity.

### 1 Search

Enter H040\_L045\_C095, 1382 or #D60000.

### 2 Inspect

Confirm the reference fields and active master provenance.

### 3 Add

Choose Add to palette. Duplicate references are blocked.

## PALETTE Manage My Palettes

- ✓ Create, rename, duplicate or delete local palettes.
- ✓ Each palette can hold up to 64 references; the workspace can hold up to 50 palettes.
- ✓ Reorder colours with arrows or drag and drop.
- ✓ Export Clarus JSON after meaningful changes. Local browser storage is not a backup.

## EXPORT Exports

| Export       | Use                                                                     |
|--------------|-------------------------------------------------------------------------|
| Clarus JSON  | Portable identity and provenance backup. Preferred for restoration.     |
| ASE          | Exact master-RGB swatches for Adobe and Affinity; verified by readback. |
| GPL          | Palette exchange for GIMP and Inkscape.                                 |
| Figma Tokens | Figma-compatible token JSON.                                            |
| CSS          | CSS custom properties with ATLAS naming and provenance.                 |

## 05 Explore without changing identity

The Wheel visualises relationships around the selected reference. Its sRGB screen context never replaces the master identity.

- ✓ Review approximate white and dark text appearance.
- ✓ Review illumination resilience metadata.
- ✓ Lock one reference for ARBE  $\lambda^*$  comparison.
- ✓ Explore complementary, analogous and triadic relationships.
- ✓ Review the tonal family with the same hue and chroma across  $L^*$ .

**Accessibility.** The screen contrast preview is approximate and is not medical or regulatory certification.

### QC Append measured QC

#### 1 Select the target

The frozen ATLAS identity remains immutable.

#### 2 Measure the sample

Use a documented instrument and measurement condition outside the bundle.

#### 3 Enter the data

Enter  $L^*$ ,  $a^*$ ,  $b^*$  and the instrument or source.

#### 4 Evaluate

RC22 reports  $\Delta E^*_{ab}$  for local review. It is not  $\Delta E_{00}$  and is not stored in the master.

## 06 Prepare independent 4C and ECG paths

Both paths start from the same frozen reference or palette. Device values from one path are never passed into the other.

### 1 Name the job

Use a project and revision-specific name.

### 2 Choose the source

Use selected reference, active palette or imported handoff JSON.

### 3 Attach the 4C profile

Upload a printer-class CMYK ICC profile, no larger than 16 MiB.

### 4 Attach the ECG profile

Upload a printer-class 7CLR CMYKOGV profile. A CMYK profile cannot be used here.

### 5 Record production metadata

Enter printing condition and substrate. Choose intent and BPC independently.

### 6 Create profile-bound data

Run the enabled calculation. Keep NOT\_MEASURED until real measurement.

### 7 Export evidence

Export both paths as JSON and the readable preparation report.

| Setting            | Expected value                                                              |
|--------------------|-----------------------------------------------------------------------------|
| 4C profile         | Printer profile with CMYK device space.                                     |
| ECG profile        | Printer profile with 7CLR CMYKOGV device space.                             |
| Rendering intent   | Perceptual, Relative colorimetric, Saturation or Absolute colorimetric.     |
| BPC                | On or Off, recorded separately for each path.                               |
| Printing condition | For example PSO Coated v3 or FOGRA55 CMYKOGV exchange colour space TAC 300. |
| Substrate          | The actual intended paper, board, film, textile or other substrate.         |

**Compatibility.** Image preview requires profiles with both A2B and B2A transforms. Header-only or one-way files are rejected.

## 07 Create parallel image previews

The preview module performs independent local LittleCMS round trips. The source is separated into the selected device space and returned to sRGB for screen comparison.

### 1 Load an image

Load PNG, JPEG or WebP, or reuse the Image Picker image. Longest side is limited to 1200 px.

### 2 Confirm both profiles

Create both previews is available when compatible profiles are attached.

### 3 Run

Create both together or run only the required path.

### 4 Export

Save each before-and-after PNG and preview-details JSON.

ATLAS CLARUS

# Colour identity before production.

One source. One frozen PKL reference. Two production previews.

Browser-sRGB → PKL FULL REFERENCE → 4C / 7C ECG



## REFERENCE

PKL Full Reference  
Frozen identity



## 4C PREVIEW

PSOcoated\_v3  
ICC round-trip simulation



## 7C ECG PREVIEW

FOGRA55 CMYKGV  
ICC round-trip simulation

## DIGITAL COMPARISON

**56%** lower mean pixel deviation

**45%** lower RMSE for ECG vs. tested 4C preview

## FOR START-UPS · DESIGN STUDIOS · PRINTERS · TECHNOLOGY PARTNERS

Make the intended colour traceable before suppliers and devices reinterpret it.

ICC PREVIEW · NOT A PHYSICAL PRINT MEASUREMENT · NOT A CERTIFIED PROOF

**EVERY COLOUR HAS AN ADDRESS.**

*Example: one PKL reference compared with tested 4C and seven-channel ECG previews. Percentages describe this digital test only.*

**Do not claim.** A round-trip preview is not paper-white simulation, a physical print, contract proof, certification or production approval.

# 08 Create reference cards and evidence

## CARD Reference card

A reference card documents the frozen identity. Its displayed or printed appearance does not modify the PKL master.

- ✓ Keep ATLAS address and atlas\_row\_id visible.
- ✓ Include master provenance and creation metadata.
- ✓ Treat an unmeasured card as NOT\_MEASURED.
- ✓ Preflight any production PDF for colour spaces, output intent, fonts and separations.

## FILES Evidence files

| File               | Proves                                    | Cannot prove                     |
|--------------------|-------------------------------------------|----------------------------------|
| Clarus JSON        | Identity and master provenance.           | Physical appearance.             |
| Print handoff JSON | Profiles, settings and independent paths. | Printer compliance.              |
| Readable report    | Human-readable preparation.               | Colour approval.                 |
| B/A PNG            | Screen comparison.                        | Measured paper result.           |
| Preview JSON       | Profile hash, settings and method.        | Press calibration.               |
| Device PDF         | Encoded values and output intent.         | Correct plates before preflight. |

**Archive.** Keep the Clarus JSON, handoff JSON, report, profile evidence, production PDF, preflight report and later measurement together.

# 09 Appearance and app connections

Appearance Pixel Simulator illustrates material, finish, illumination, viewing angle and surface depth. It is not a spectral renderer or physical-material prediction.

| Control       | Examples                                                                      |
|---------------|-------------------------------------------------------------------------------|
| Material      | Coated paperboard, polymer film, brushed metal, woven textile, glass or wood. |
| Finish        | A visual finish option provided by the interface.                             |
| Illumination  | A simulated viewing-light context.                                            |
| Viewing angle | An illustrative change in geometry.                                           |
| Surface depth | A visual depth treatment.                                                     |

## APPS External applications

- ✓ Adobe and Affinity: import ASE and retain the ATLAS reference as swatch name.
- ✓ GIMP and Inkscape: import GPL and preserve Clarus JSON as evidence.
- ✓ Figma: use token JSON; its RGB is the documented screen representation.
- ✓ CSS: use generated custom properties and keep provenance comments.
- ✓ Krita and Scribus: follow included demonstrations and keep identity separate from colour conversion.

# 10 Troubleshooting

| Problem                    | Cause                                                           | Action                                                                       |
|----------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------|
| Application fails to open. | ZIP not extracted or files moved.                               | Extract the complete archive again and open index.html in the intact folder. |
| Palette disappeared.       | Browser storage cleared or file moved.                          | Import the last Clarus JSON backup.                                          |
| ICC profile rejected.      | Wrong device space, invalid header, size or missing transforms. | Use CMYK for 4C or 7CLR CMYKOGV for ECG, below 16 MiB.                       |
| Preview button disabled.   | Image or compatible profile missing.                            | Load an image and attach the required profile or both profiles.              |
| Preview differs strongly.  | Output gamut and settings differ.                               | Review profile, intent, BPC and condition; do not alter identity to hide it. |
| Physical print differs.    | Press, substrate, calibration, viewing or conversion changed.   | Preflight, document the RIP path and measure the dried print.                |

## CHECK Before printer handoff

- ✓ Confirm ATLAS address, atlas\_row\_id and master SHA-256.
- ✓ Confirm the exact ICC profile and its SHA-256.
- ✓ Record printing condition, substrate, rendering intent and BPC.
- ✓ Preflight the production PDF and inspect all intended separations.
- ✓ Tell the printer not to apply an unintended additional conversion.
- ✓ Print under the agreed condition and measure the dried sample.

# 11 Version and claim record

| Item            | RC22 value                                                       |
|-----------------|------------------------------------------------------------------|
| Bundle          | ATLAS Clarus Browser Bundle v0.2.0 rc22 parallel image preview   |
| Archive checked | rc22.zip                                                         |
| Archive SHA-256 | 2adc5120a98b6338c4a7b6e247712a3347424c03e3a14cb9fda3bf40148350ad |
| Reference count | 13,283                                                           |
| Master SHA-256  | 8283ab91b10f89ac758d09ecf5fb4d6343536600a06dd468b1cc1ecf4ec747c4 |
| Validation      | READY_PENDING_VISUAL_AUDIT                                       |
| Preview engine  | LittleCMS 2.16 via lcms-wasm 1.0.5                               |
| Method          | sRGB to device 16-bit to sRGB                                    |
| Measurement     | NOT_MEASURED unless a real record is appended                    |
| A' logic        | v0.4 unchanged; v0.5 diagnostics do not change selection         |

## CAUTION Claim boundary

ATLAS Clarus identifies, freezes and documents a reference colour and its preparation paths. It does not guarantee that every display, printer, ink set or material can reproduce that colour. ICC previews are computational simulations. Physical claims require a controlled print, documented viewing conditions and measurement.

## CREDITS Credits and licensing

Review LICENSING.html, LCMS-LICENSE.txt, LCMS-WASM-LICENSE.txt and LCMS-PROVENANCE.json in the bundle before redistribution. Credit is due to freieFarbe e. V. and freieFarbe.de for the open HLC colour reference and foundational work.

**End of manual**